

Circular Insecure Encryption: from Long Cycles to Short Cycles

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Abstract. A length n encryption cycle consists of a sequence of n keys, each encrypting the next, forming a cycle, and an encryption scheme is n -circular secure if a length n encryption cycle is computationally indistinguishable from encryptions of zeros. An interesting problem is whether CPA security implies circular security. This is shown to be not true. Using standard cryptographic assumptions and LWE, it was shown that within the class of CPA secure encryption schemes, for any n , there exists an n -circular insecure encryption scheme. Furthermore, there exists a particular encryption scheme that is ℓ -circular insecure for all ℓ . Following these results, it is natural to ask whether a circular insecurity of a particular length implies circular insecurity of different lengths and of multiple lengths. We answer this problem with an affirmative in this paper. We constructively prove that a CPA secure encryption scheme that is insecure in the presence of encryption cycles of length $(n + 1)$ implies the existence of such a scheme for encryption cycles of any length less than $(n + 1)$. The constructed $(\leq n)$ -circular insecure construction may have the same message space as the $(n + 1)$ -circular insecure encryption scheme, and our results apply to both public key and symmetric key settings.

Keywords: Encryption Cycle · Circular Security · CPA

1 Introduction

The question of whether encryption security is maintained in the presence of self-encryption was identified during the early stages of modern cryptography [GM84], and in general remains unanswered. This was later generalized, leading to the study of *circular security*. Informally, n -circular security asks for the following two sequences of ciphertexts to be indistinguishable,

$$\begin{aligned} & \text{enc}(\text{pk}_0, \text{sk}_1), \text{enc}(\text{pk}_1, \text{sk}_2), \dots, \text{enc}(\text{pk}_{n-1}, \text{sk}_0) \\ & \text{enc}(\text{pk}_0, 0^{|\text{sk}_1|}), \text{enc}(\text{pk}_1, 0^{|\text{sk}_2|}), \dots, \text{enc}(\text{pk}_{n-1}, 0^{|\text{sk}_0|}). \end{aligned}$$

Theoretical considerations of *circular security* arise in a number of areas. One is the study of symbolic (formal) encryption [DY83], in which formal expressions over atomic sets of keys and data represent an idealized encryption functionality, and adversarial knowledge is modeled via an entailment relation over expressions. Abadi and Rogaway [AR00] sought to reconcile this symbolic approach with the computational approach, thereby highlighting the gap between them introduced by so-called circular encryption. It was thought best to avoid circular encryption in the formal setting in order to achieve soundness of their formal model with respect to CPA security. This gap is further studied in [Lau02, CZ06, ABHS09, Mic19, MW04, Mic10, HK13, MP05]. Without additional constraints on the symbolic side, circular security is also crucial to the completeness of a

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logic. Another area in which circular security has become significant is the construction of various primitives whose security relies on underlying primitives with circular security properties. For example, circular security assumption is used to construct homomorphic encryption [Gen09, GSW13, BV14] and indistinguishability obfuscation [GP21]. A well-known example of how circular encryption arises in practice is Windows BitLocker [Mic24], where a key may be used to encrypt a disk on which the key is stored as data. Not only is circular security useful, but applications with provable circular insecurity are also emerging. Using certain provable circular insecurity with other assumptions, a lockable obfuscation is constructed in [Klu22], and a witness encryption scheme is constructed in [Klu20]. We refer to provable circular insecurity because these constructions explicitly use cycle testers.

Given its potential for application in a variety of areas, there have been numerous works dealing with the (im)possibility of circular security under different scenarios, [App14, KRW15, ABBC10, BHHO08, AP16, BHW15, HK17, CGH12, Rot13, GH10, GKW17c, MO14, KM20, BHH10, BGK11, WZ17]. A particularly intriguing line of work involves finding relations between various key-cycle lengths and message spaces for circular secure schemes [HK17, Rot13, KM20, KRW15]. Recently, it was shown in [WW23] that Key Dependent Message (KDM) security, a more demanding notion than that of circular security [BRS03], is reducible to 1-circular security of bit encryption, which is when an encryption scheme that encrypts messages one bit at a time is secure under the presence of length 1 encryption cycles.

Motivated by these questions, we are interested in finding other relations between notions of circular security. In this paper, we show that a longer-length circular insecure encryption scheme implies the existence of a shorter-length circular insecure encryption scheme.

1.1 Related Work and Motivation

Relations between previously defined security notions, key/message length, and key-cycle length have been widely studied.

Folklore Construction. When the message length is larger than or equal to the secret key length, folklore constructions exist for CPA secure encryptions that are 1-circular secure and 1-circular insecure (see Appendix A). The core idea of the folklore constructions is to add a step in the encryption function to determine whether the message is equal to the secret key. Or as others have phrased it, “public key pk can ‘communicate’ to see if m is pk ’s secret key.” [HK17] Our version of the folklore construction is as follows. Using a base CPA secure encryption scheme and a one-way function f , we can construct a new encryption scheme where the new key generation function first calls the old generation function twice to receive two public-secret key pairs $(\text{pk}_0, \text{sk}_0), (\text{pk}_1, \text{sk}_1)$. We then throw away pk_1 , and output $(f(\text{sk}_1), \text{pk}_0)$ and $(\text{sk}_0, \text{sk}_1)$ as the new public key and secret key respectively. The new encryption function with message $m = (m[0], m[1])$ under the scenario of $f(m[1]) = f(\text{sk}_1)$ ¹, may leak information about the message to give 1-circular insecurity or encrypt a random string to give 1-circular security.

Cycles with Arbitrary Lengths. However, for encryption cycles of longer lengths, known results demonstrate that it is harder to find instances of secure or insecure schemes and these often require specific assumptions such as DDH for circular security [BHHO08], and SXDH, indistinguishability obfuscation, or LWE for circular insecurity [Rot13, KRW15, BHW15, AP16, CGH12, GKW17a, GKW17b, GKW17c]. Given the simplicity of the folklore constructions, it is desirable to determine implications between n -circular (in)secure encryption schemes and $(n + 1)$ -circular (in)secure encryption schemes. In [HK17], it was shown that going from an n -circular secure encryption to an $(n + 1)$ -circular secure

¹To the best of our knowledge, all versions of the folklore construction on 1-circular secure scheme do not satisfy the strict definition of correctness.

encryption scheme using a black-box construction is likely to be difficult. Using a drastically different method, [WW23] showed that the existence of a 1-circular secure symmetric key bit encryption scheme implies the existence of an n -circular (n -key KDM) and CPA secure encryption scheme. In light of this result, the search for an n -circular secure encryption scheme becomes a search for a 1-circular secure bit encryption scheme, which is also shown to be difficult in the black-box setting by [HK17, Rot13].

Motivation. In [KRW15], it was shown that using indistinguishability obfuscation, for any n , an n -circular insecure and CPA secure encryption scheme can be constructed. Furthermore, in [GKW17b], using indistinguishability obfuscation and leveled fully homomorphic encryption, a CPA secure encryption scheme that is not circular secure for all lengths is presented. Although indistinguishability obfuscation was used to achieve these results, it is shown in [GKW17a, WZ17] that lockable obfuscation suffices, which can be constructed from LWE along with the standard cryptographic assumption. These results inspire the following two questions in the CPA setting. The first question is whether a circular insecure encryption scheme of a particular length can be used to construct an encryption scheme that is circular insecure of a different length, in particular:

1. Does n -circular insecurity imply $(n + 1)$ -circular insecurity?
2. Does $(n + 1)$ -circular insecurity imply n -circular insecurity?

The second question is whether a circular insecure encryption scheme of a particular length can be used to construct an encryption scheme that is circular insecure of multiple lengths. We answer 2. of the first question and the second question with an affirmative answer by constructing a $(\leq n)$ -circular insecurity encryption scheme from a $(n + 1)$ -circular insecurity encryption scheme. Our work provides a novel relationship between the counterexamples.

Prior works that show counterexamples for 1-circular bit encryption involve specific assumptions such as LWE, multilinear maps, or indistinguishability obfuscation [GKW17c, Rot13, KRW15]. Our construction 2 provides a way to build 1-circular insecure bit encryption from any n -circular insecure bit encryption scheme. This is a result not offered in prior works.

Our result can be used to simplify scenarios where circular insecurity of multiple lengths is needed. For example, consider the following sequence of ciphertexts:

$$\varphi = \text{enc}(\text{pk}_1, \text{enc}(\text{pk}_0, \text{sk}_0)), \text{enc}(\text{pk}_2, \text{sk}_1), \text{enc}(\text{pk}_2, \text{sk}_3), \text{enc}(\text{pk}_3, \text{sk}_2)$$

Suppose the task is to find a CPA secure encryption scheme Π with encryption function enc and key generation function KeyGen where $(\text{pk}_i, \text{sk}_i) \leftarrow \text{KeyGen}(1^\eta)$ for $i \in \{0, \dots, 3\}$ such that an adversary can recover sk_0 from φ . If there exists an encryption scheme where an adversary can recover a key from a length 3 encryption cycle, then we can construct a scheme Π such that the keys of its length 2 and length 1 encryption cycles are recoverable. This implies that sk_0 is recoverable from φ in Π .

1.2 Our Contribution

We present new implications for circular (in)security. To be more precise, we show that if an $(n + 1)$ -circular insecure CPA secure encryption scheme exists, then a $(\leq n)$ -circular insecure CPA secure encryption scheme exists. This result applies to both public key and symmetric key encryption schemes. We present the proof for the public key setting, which, with slight modification, can serve as a proof in the symmetric key setting.

The proof uses a circular insecure encryption scheme as a black-box to construct another encryption scheme. In Construction 1, we present a construction that does not preserve the message space of the initial encryption scheme. In Construction 2, we construct an encryption scheme that preserves the message space of the base encryption scheme if

the message space of the base scheme consists of only one bit. The techniques used in Construction 2 can be modified to preserve message spaces for various base schemes. The motivation for these two specific constructions is that Construction 1 is very intuitive, and a single-bit message space is frequently considered in the study of circular security.

Intuitively, the idea is to use an $n + 1$ -circular insecure encryption scheme Π to create an encryption scheme Π' so that a key of Π' consists of multiple keys in Π . This is done in a way so that from any length ℓ encryption cycle where $0 < \ell < n + 1$ of Π' , we can extract an encryption cycle with length $n + 1$ of Π . We demonstrate this intuition by considering an encryption scheme Π with the following length 2 encryption cycle,

$$\text{enc}(\text{pk}_0, \text{sk}_1), \text{enc}(\text{pk}_1, \text{sk}_0).$$

We define a function f with two parameters in the following way:

$$f((\varphi_0, \varphi_1, \mathbf{s}), \psi) = (\text{enc}(\varphi_1, \psi), \mathbf{s}).$$

Let f be the encryption function for a scheme Π' where a public and a secret key takes the form $(\text{pk}_0, \text{pk}_1, \mathcal{E}(\text{pk}_0, \text{sk}_1))$ and sk_0 respectively. We observe that we can find a length 2 encryption cycle of Π in a length 1 cycle of Π' :

$$f((\text{pk}_0, \text{pk}_1, \text{enc}(\text{pk}_0, \text{sk}_1)), \text{sk}_0) = (\text{enc}(\text{pk}_1, \text{sk}_0), \text{enc}(\text{pk}_0, \text{sk}_1)).$$

The “additional” information $\mathbf{s}$ appended to the key does not reveal anything when not forming a length 2 cycle of Π , and that is unlikely to occur unless f is used to create a length 1 cycle. This guarantee is provided by the CPA security of the base encryption scheme Π .

1.3 Future Work

There are two interesting directions that we would like to pursue. The first is to show the converse of our result. Although it has been shown to be difficult to construct an $(n + 1)$ -circular secure encryption scheme from an n -circular secure encryption scheme in the black-box setting [HK17], constructing $(n + 1)$ -circular insecurity from n -circular insecurity has not been shown to be difficult. The second is designing an encryption scheme that is both circular secure and circular insecure for some specified lengths. Combining these two ideas may lead to interesting protocols.

Organization.

We will present the notations and definitions in Section 2. In Section 3, we present Construction 1, a message length non-preserving black-box construction. Its analysis follows in Section 4. Section 5 contains Construction 2, in which we discuss preserving a message space of only one bit in the black-box technique. The section ends with a generalization of Construction 2 to preserve arbitrary message space. We present a concise discussion of how to modify the results of this paper to the symmetric key setting in Section 6.

2 Preliminaries

We use PPT to denote probabilistic polynomial time. For a set S , we write $s \leftarrow \$S$ to indicate that s is sampled uniformly from S . Flipping a coin x means $x \leftarrow \$\{0, 1\}$. For a randomized function f with input x , we write $y \leftarrow f(x)$ to indicate that y is the output of $f(x)$ with fresh randomness. We write $\langle \dots \rangle$ for a tuple. For characters 0, 1, and a number variable x , we write 0^x or 1^x to denote the unary string with 0 or 1 repeated x a number

of times. For a string s , we write $|s|$ to denote the length of s . For two strings s_0 and s_1 , we write $s_0 \| s_1$ to denote the concatenation of s_0 and s_1 . For variables, both the subscript and the superscript are used to label different variables.

Definition 1. A *public key encryption scheme* consists of three efficiently computable randomized functions: a key generating function $\mathcal{G}$, encryption function $\mathcal{E}$, and decryption function $\mathcal{D}$.

- The key generating function $\mathcal{G}$ takes a unary string 1^η , where η is the security parameter and outputs a pair, (pk, sk) , consisting of a public key and a secret key. We assume for any $(\text{pk}_i, \text{sk}_i) \leftarrow \mathcal{G}(1^\eta)$ and $(\text{pk}_j, \text{sk}_j) \leftarrow \mathcal{G}(1^\eta)$ that $|\text{pk}_i| = |\text{pk}_j|$ and $|\text{sk}_i| = |\text{sk}_j|$.
- The encryption function $\mathcal{E}$ takes a public key pk and a message m and outputs a ciphertext c .
- The decryption function $\mathcal{D}$ takes a secret key sk and a ciphertext c and outputs a message m or $\perp$.

As usual, we require ² that for any $(\text{pk}, \text{sk}) \leftarrow \mathcal{G}(1^\eta)$ and any message m , $\mathcal{D}(\text{sk}, \mathcal{E}(\text{pk}, m)) = m$.

A (public key) bit encryption scheme has message space $\{0, 1\}$. For a bit encryption scheme and a message m where $|m| > 1$, we write $\mathcal{E}(\text{pk}, m)$ to denote that bit-by-bit encryption of m using $\mathcal{E}(\text{pk}, \cdot)$, each encryption using fresh randomness. More definitions of bit encryption in circular settings can be found [Rot13].

The definition of a CPA experiment and CPA security follows that of [KL07].

Definition 2. Let $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$ be a public key encryption scheme. The CPA experiment for Π and adversary $\mathcal{A}$ denoted by $\text{CPA}_{\mathcal{A}, \Pi}(\eta)$ is defined as follows:

$\text{CPA}_{\mathcal{A}, \Pi}(\eta) :$

1. $\text{pk}, \text{sk} \leftarrow \mathcal{G}(1^\eta)$. Give η and pk to $\mathcal{A}$.
2. $b \leftarrow \$ \{0, 1\}$.
3. $\mathcal{A}$ outputs two equal length challenge messages m_0, m_1 .
4. Compute $c^b \leftarrow \mathcal{E}(\text{pk}, m_b)$ and return c^b to $\mathcal{A}$.
5. $\mathcal{A}$ outputs b' and the experiment results in 1 if $b = b'$, 0 otherwise.

We say Π is CPA secure if for every PPT adversary $\mathcal{A}$, for some negligible function negl ,

$$\Pr[\text{CPA}_{\mathcal{A}, \Pi}(\eta) = 1] \leq \frac{1}{2} + \text{negl}(\eta).$$

The definition of n -circular experiment and n -circular-security follows that of [CGH12].

Definition 3. Let $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$ be a public key encryption scheme. The n -circular experiment for Π and adversary $\mathcal{A}$ denoted by $\text{CIRC}_{\mathcal{A}, \Pi}(n, \eta)$ is defined as follows:

$\text{CIRC}_{\mathcal{A}, \Pi}(n, \eta) :$

1. $\text{pk}_0, \text{sk}_0 \leftarrow \mathcal{G}(1^\eta), \dots, \text{pk}_{n-1}, \text{sk}_{n-1} \leftarrow \mathcal{G}(1^\eta)$. Give η and $\text{pk}_0, \dots, \text{pk}_{n-1}$ to $\mathcal{A}$.
2. $b \leftarrow \$ \{0, 1\}$.

²This is often required except with negligible probability over the key pairs [KL07]. This does not change any of our results.

3. Compute for $i \in \{0, \dots, n-1\}$:

$$c_i^b \leftarrow \begin{cases} \mathcal{E}(\text{pk}_i, \text{sk}_{i+1 \bmod n}) & \text{If } b = 1 \\ \mathcal{E}(\text{pk}_i, 0^{|\text{sk}_1|}) & \text{If } b = 0 \end{cases}$$

Send $c_0^b, \dots, c_{n-1}^b$ to $\mathcal{A}$.

4. When $\mathcal{A}$ outputs b' , the experiment results in 1 if $b = b'$, 0 otherwise.

We say Π is n -circular secure if for every PPT adversary $\mathcal{A}$, for some negligible function negl ,

$$\Pr[\text{CIRC}_{\mathcal{A}, \Pi}(n, \eta) = 1] \leq \frac{1}{2} + \text{negl}(\eta).$$

An encryption scheme is n -circular insecure if it is not n -circular secure. An encryption scheme is $(\leq n)$ -circular insecure if it is ℓ -circular insecure for any $\ell \in \{1, \dots, n\}$.

Although it is unknown whether CPA security implies circular security or not, CPA security provides indistinguishability on ciphertexts that are “almost” an encryption cycle. What we mean by this is the following: consider a sequence of ciphertexts that form an encryption cycle, if we replace some of these ciphertexts with encryptions of zeros, then this sequence is indistinguishable from the sequence of ciphertexts that are all encryptions of zeros with the same keys. We denote this experiment between an adversary $\mathcal{A}$ and an encryption scheme Π as $\text{ZERO}_{\mathcal{A}, \Pi}(n, t, \eta)$. It is required that $n, t \geq 1$, and η is the security parameter. The parameter n is the length of an encryption cycle or just the number of ciphertexts in a sequence when encrypting zeros. In the experiment, the first t ciphertexts of an encryption cycle are replaced with encryptions of zero.

$\text{ZERO}_{\mathcal{A}, \Pi}(n, t, \eta)$:

1. $(\text{pk}_0, \text{sk}_0) \leftarrow \mathcal{G}(1^\eta), \dots, (\text{pk}_{n-1}, \text{sk}_{n-1}) \leftarrow \mathcal{G}(1^\eta)$. Send η and $\text{pk}_0, \dots, \text{pk}_{n-1}$ to $\mathcal{A}$.
2. A random coin b is flipped.
3. For $i \in \{0, \dots, n-1\}$ compute

$$c_i^b \leftarrow \begin{cases} \mathcal{E}(\text{pk}_i, \text{sk}_{i+1 \bmod n}) & \text{if } b = 1 \text{ and } i \geq t, \\ \mathcal{E}(\text{pk}_i, 0^{|\text{sk}_1|}) & \text{otherwise.} \end{cases}$$

Send $c_0^b, \dots, c_{n-1}^b$ to $\mathcal{A}$.

4. When $\mathcal{A}$ outputs b' , the experiment result in 1 if $b = b'$, 0 otherwise.

Lemma 1. *If a public key encryption scheme $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$ is CPA secure. Then for any PPT adversary $\mathcal{A}$, any $n \geq 1$, and any $t \geq 1$,*

$$\Pr[\text{ZERO}_{\mathcal{A}, \Pi}(n, t, \eta) = 1] \leq \frac{1}{2} + \text{negl}(\eta)$$

for all but finitely many η and some negligible function negl .

Applying the soundness result from [Mic10] is an easy method to show this. However, we provide the following proof to keep this paper self-contained without elaborating on formal cryptography.

Proof. Assume that Π is CPA secure. We will show that for any PPT adversary $\mathcal{A}$, and some negligible function negl ,

$$|\Pr[\text{ZERO}_{\mathcal{A}, \Pi}(n, t, \eta) = 1] - \Pr[\text{ZERO}_{\mathcal{A}, \Pi}(n, t+1, \eta) = 1]| \leq \text{negl}(\eta)$$

for any $n, t \geq 1$. The statement then follows, since n is not dependent on η and for $t' \geq n$ it must be the case that

$$\Pr[\text{ZERO}_{\mathcal{A}, \Pi}(n, t', \eta) = 1] = \frac{1}{2}.$$

This is because when $t' \geq n$, the adversary is receiving encryptions of zeros in the experiment regardless of the value of b .

Consider an arbitrary adversary $\mathcal{A}$ such that

$$|\Pr[\text{ZERO}_{\mathcal{A}, \Pi}(n, t, \eta) = 1] - \Pr[\text{ZERO}_{\mathcal{A}, \Pi}(n, t+1, \eta) = 1]| = \frac{1}{2} + \epsilon(\eta).$$

Now consider an adversary $\mathcal{A}'$ to play in $\text{CPA}_{\mathcal{A}', \Pi}(\eta)$ defined as follows.

$\mathcal{A}'$:

1. Receive η and pk . Label this pk as pk_t .
2. For $i \in \{0, \dots, n-1\} \setminus \{t\}$ compute $(\text{pk}_i, \text{sk}_i) \leftarrow \mathcal{G}(1^\eta)$. Give $\text{pk}_0, \dots, \text{pk}_{n-1}$ to $\mathcal{A}$.
3. Query challenge messages $m_0 = 0^{|\text{sk}|}$ and $m_1 = \text{sk}_{t+1}$ to receive $c_t^b \leftarrow \mathcal{E}(\text{pk}_t, m_b)$.
4. For $i \in \{0, \dots, n-1\} \setminus \{t\}$ compute

$$c_i \leftarrow \begin{cases} \mathcal{E}(\text{pk}_i, 0^{|\text{sk}|}) & \text{if } i < t \\ \mathcal{E}(\text{pk}_i, \text{sk}_{i+1 \bmod n}) & \text{otherwise.} \end{cases}$$

Give $c_0, \dots, c_t^b, \dots, c_n$ to $\mathcal{A}$.

5. When $\mathcal{A}$ outputs b' , output b' .

It can be observed that when $b = 0$, $\mathcal{A}'$ has simulated $\text{ZERO}_{\mathcal{A}, \Pi}(n, t+1, \eta)$ and when $b = 1$, $\mathcal{A}'$ has simulated $\text{ZERO}_{\mathcal{A}, \Pi}(n, t, \eta)$. Therefore, we conclude that if $\epsilon(\eta)$ is not negligible, then Π is not CPA secure. $\square$

3 Construction of n-Circular Insecure Encryption

We show how to construct a CPA secure ($\leq n$)-circular insecure public key encryption from a CPA secure $(n+1)$ -circular insecure public key encryption. We present Construction 1 in this section, which does not preserve message space. Our Construction 2, which preserves the message space of the bit encryption scheme, will be presented in Section 5.

Construction 1. *Given a CPA secure $(n+1)$ -circular insecure public key encryption scheme $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$. We obtain $\Pi' = (\mathcal{G}', \mathcal{E}', \mathcal{D}')$ as follows:*

Π' :

- $\mathcal{G}'(1^\eta)$: Compute

$$(\text{pk}[0], \text{sk}[0]) \leftarrow \mathcal{G}(1^\eta), \dots, (\text{pk}[n], \text{sk}[n]) \leftarrow \mathcal{G}(1^\eta).$$

Compute $\mathbf{s} \leftarrow \mathcal{E}(\text{pk}[0], \text{sk}[n])$. For $3 \leq i \leq n$ compute $h[i] \leftarrow \mathcal{E}(\text{pk}[i], \text{sk}[i-1])$.
Return public key

$$\text{pk} = \text{pk}[0], \dots, \text{pk}[n], \mathbf{s}, h[n], h[n-1], \dots, h[3].$$

and secret key

$$\text{sk} = \text{sk}[0], \dots, \text{sk}[n-1].$$

We note that when $n < 3$, the public key has nothing after $\mathbf{s}$.

- $\mathcal{E}'(\text{pk}, m)$: Parse pk to

$$\text{pk}[0], \dots, \text{pk}[n], s, h[n], \dots, h[3].$$

If $|m| \neq |sk|$, compute $c \leftarrow \mathcal{E}(\text{pk}[1], m)$ and return

$$\langle 0, c \rangle.$$

Otherwise, parse m to $m[0], \dots, m[n-1]$ where $|m[i]| = |sk[i]|$ for $0 \leq i \leq n-1$. For $0 \leq i \leq n-1$ compute $c[i] \leftarrow \mathcal{E}(\text{pk}[i+1], m[i])$, return

$$\langle 1, c[0], \dots, c[n-1], s \rangle.$$

- $\mathcal{D}'(\text{sk}, c)$: Parse sk to

$$\text{sk}[0], \dots, \text{sk}[n-1].$$

If $c = \langle 0, c' \rangle$, return $\mathcal{D}(\text{sk}[1], c')$. Else parse c into $\langle 1, c[0], \dots, c[n-1], s \rangle$, compute $\text{sk}[n] \leftarrow \mathcal{D}(\text{sk}[0], s)$ and return

$$\mathcal{D}(\text{sk}[1], c[0]) \parallel \dots \parallel \mathcal{D}(\text{sk}[n], c[n-1]).$$

One can observe that if all functions of Π are efficiently computable, so are the functions of Π' . Similarly, one can observe that if Π is correct, then so is Π' .

Remark 1. We can delete the terms $h[\dots]$ to achieve an $(n+1)$ -circular insecurity to n -circular insecurity construction. Such a construction has reduced key size, and although the theoretical implications of our result are unchanged, it raised concerns about the key size blowup of going from $(n+1)$ -circular insecurity to 1-circular insecurity.

We provide the Figures 1 and 2 to visually represent how a length 3 encryption cycle of Π can be found in a length 1 and length 2 encryption cycle of Π' when $n = 2$. These examples are given in detail in Appendix C. These examples aim to demonstrate the intuition behind Construction 1. Although the terms $h[\dots]$ are not used in this example, they are used in the example of subsection 4.1, where we set $n = 3$. The goal of subsection 4.1 is to illustrate the method that we use in the proof of Lemma 2.

Figure 1 visually represents how to spot a length 3 cycle of Π in a length 1 cycle of Π' , which is $\mathcal{E}'(\text{pk}, \text{sk})$ for any $(\text{pk}, \text{sk}) \leftarrow \mathcal{G}'(1^n)$. We interpret this graph by first observing that

$$\mathcal{E}'(\text{pk}, \text{sk}) = \langle 1, \mathcal{E}(\text{pk}[1], \text{sk}[0]), \mathcal{E}(\text{pk}[2], \text{sk}[1]), s \rangle \text{ and } s = \mathcal{E}(\text{pk}[0], \text{sk}[2]).$$

where $\text{pk} = \text{pk}[0], \text{pk}[1], \text{pk}[2], s$ and $\text{sk} = \text{sk}[0], \text{sk}[1]$. We will separate each element of this tuple with boxes. Then, following the directed arrows in the figure, each ciphertext is an encryption of the secret key corresponding to the public key used to make the next ciphertext.

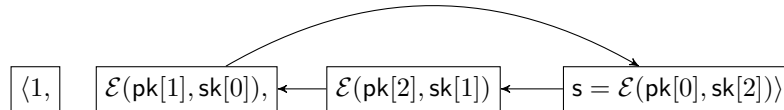


Figure 1: 3-cycle of Π in 1-cycle of Π'

Figure 2 visually represents how to spot a length 3 cycle of Π in a length 2 cycle of Π' , which is $\mathcal{E}'(\text{pk}_0, \text{sk}_1), \mathcal{E}'(\text{pk}_1, \text{sk}_0)$ given $(\text{pk}_0, \text{sk}_0) \leftarrow \mathcal{G}'(1^n)$ and $(\text{pk}_1, \text{sk}_1) \leftarrow \mathcal{G}'(1^n)$. In the figure, we have

$$\mathcal{E}'(\text{pk}_0, \text{sk}_1) = \langle 1, \mathcal{E}(\text{pk}_0[1], \text{sk}_1[0]), \mathcal{E}(\text{pk}_0[2], \text{sk}_1[1]), s_0 \rangle \text{ and } s_0 = \mathcal{E}(\text{pk}_0[0], \text{sk}_0[2])$$

at the top row and

$$\mathcal{E}'(\text{pk}_1, \text{sk}_0) = \langle 1, \mathcal{E}(\text{pk}_1[1], \text{sk}_0[0]), \mathcal{E}(\text{pk}_1[2], \text{sk}_0[1]), s_1 \rangle \text{ and } s_1 = \mathcal{E}(\text{pk}_1[0], \text{sk}_1[2])$$

at the bottom row where for $i \in \{0, 1\}$, $\text{pk}_i = \text{pk}_i[0], \text{pk}_i[1], \text{pk}_i[2], s_i$ and $\text{sk}_i = \text{sk}_i[0], \text{sk}_i[1]$. Each tuple has its elements separated by boxes. Following the directed arrows, we have a length 3 encryption cycle of Π .

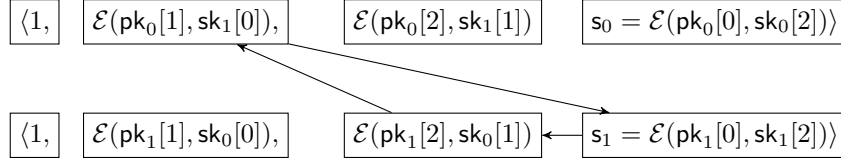


Figure 2: 3-cycle of Π in 2-cycle of Π'

We note that the cycle following the directed edge is not the only way to extract a length 3 encryption cycle of Π from the ciphertexts above.

It may be interesting to point out that for certain lengths, it is possible to construct circular security of a smaller cycle from the circular security of a longer cycle. In the figures above, if we replace Π with a 3-circular secure encryption scheme, then Figure 1 contains exactly one length 3 encryption cycle of Π . Figure 2 contains exactly two “unrelated” length 3 encryption cycles of Π . However, it is unclear how to generalize this to all integers larger than 3. But for only public key setting, and only going from $n + 1$ to n , we provide a construction in Appendix D.

Theorem 1. *If a CPA secure $(n + 1)$ -circular insecure public key encryption scheme Π exists, then Π' is a CPA secure and $(\leq n)$ -circular insecure encryption scheme.*

The following section shows this theorem via Lemma 2 and Lemma 3.

4 Analysis of Construction 1

This section will show that if Π is $(n + 1)$ -circular insecure and CPA secure, then Π' is $(\leq n)$ -circular insecure and CPA secure.

We show Π' is $(\leq n)$ -circular insecure by showing a way to find a length $n + 1$ encryption cycle of Π in a length ℓ encryption cycle of Π' , where $1 \leq \ell \leq n$. The proof of this is contained in Section 4.2, where we index the terms of the public keys and ciphertexts and show a way to find the length $n + 1$ encryption cycle of Π . We provide an example in Section 4.1 that follows such an indexing method.

The proof of CPA security of Π' is presented in Section 4.3. We will use a sequence of games to show that the advantage of an adversary winning the CPA game against Π' is identical to that of an adversary winning the CPA game against Π , except with negligible probability.

4.1 Circular Insecurity Example

In the following example, we consider the case where $n = 3$. We give an example of how to extract a length 4 encryption cycle of Π in a length 2 encryption cycle of Π' , following the method presented in the proof of Lemma 2. Let $\text{pk}_0, \text{sk}_0 \leftarrow \mathcal{G}'(1^n)$ and $\text{pk}_1, \text{sk}_1 \leftarrow \mathcal{G}'(1^n)$. For $i \in \{0, 1\}$, we have

$$\text{pk}_i = \text{pk}_i[0], \text{pk}_i[1], \text{pk}_i[2], \text{pk}_i[3], s_i, h_i[3]$$

and

$$\mathbf{sk}_i = \mathbf{sk}_i[0], \mathbf{sk}_i[1], \mathbf{sk}_i[2].$$

Given the ciphertexts $c_0 = \mathcal{E}'(\mathbf{pk}_0, \mathbf{sk}_1)$ and $c[1] = \mathcal{E}'(\mathbf{pk}_1, \mathbf{sk}_0)$ and the public keys $\mathbf{pk}_0$ and $\mathbf{pk}_1$. For $i \in \{0, 1\}$, let

$$c_i = \langle 1, c_i[0], c_i[1], c_i[2], \mathbf{s}_i \rangle.$$

Let $\ell = 2$ and $n = 3$. We can extract the following keys in the order described:

1. $\mathbf{pk}_{\ell-1}[0]$, which is $\mathbf{pk}_1[0]$.
2. For $\ell + 1 \leq i \leq n$ in descending order, $\mathbf{pk}_{\ell-1}[i]$. Which is $\mathbf{pk}_1[3]$.
3. For $0 \leq j \leq \ell - 1$ in ascending order, $\mathbf{pk}_{(\ell+j-1) \bmod \ell}[\ell - j]$. Which are $\mathbf{pk}_1[2]$, and $\mathbf{pk}_0[1]$.

So the keys we have extracted are $\mathbf{pk}_1[0], \mathbf{pk}_1[3], \mathbf{pk}_1[2], \mathbf{pk}_0[1]$, which are 4 keys of Π . We can also extract the following ciphertexts:

1. $\mathbf{s}_{\ell-1}$, which is $\mathbf{s}_1$
2. For $\ell + 1 \leq i \leq n$ in descending order, $h_{\ell-1}[i]$. Which is $h_1[3]$.
3. For $0 \leq j \leq \ell - 1$ in ascending order, $c_{(\ell+j-1) \bmod \ell}[\ell - j - 1]$. Which are $c_1[1]$ and $c_0[0]$.

We observe that the ciphertexts extracted are the following, which form a length 4 encryption cycle of Π with the 4 keys extracted.

$$\begin{aligned} \mathbf{s}_1 &\leftarrow \mathcal{E}(\mathbf{pk}_1[0], \mathbf{sk}_1[3]) \\ h_1[3] &\leftarrow \mathcal{E}(\mathbf{pk}_1[3], \mathbf{sk}_1[2]) \\ c_1[1] &\leftarrow \mathcal{E}(\mathbf{pk}_1[2], \mathbf{sk}_0[1]) \\ c_0[0] &\leftarrow \mathcal{E}(\mathbf{pk}_0[1], \mathbf{sk}_1[0]). \end{aligned}$$

4.2 Circular Insecurity

Lemma 2. *If Π is $(n + 1)$ -circular insecure and CPA secure, then Π' is $(\leq n)$ -circular insecure.*

Proof. Let Π be $(n + 1)$ -circular insecure and CPA secure. Then, there exists an adversary $\mathcal{A}$ such that

$$\Pr[\text{CIRC}_{\mathcal{A}, \Pi}(n + 1, \eta) = 1] \geq \frac{1}{2} + \epsilon(\eta).$$

Consider an arbitrary $\ell \in \{1, \dots, n\}$. We can define an adversary $\mathcal{A}'_\ell$ as follows.

$\mathcal{A}'_\ell$:

1. Receive η and $\mathbf{pk}_0, \dots, \mathbf{pk}_{\ell-1}$ where for $0 \leq x \leq \ell - 1$

$$\mathbf{pk}_x = \mathbf{pk}_x[0], \dots, \mathbf{pk}_x[n], \mathbf{s}_x, h_x[n], \dots, h_x[3].$$

Send the following keys to $\mathcal{A}$ in the order described:

- (a) $\mathbf{pk}_{\ell-1}[0]$
- (b) for $\ell + 1 \leq i \leq n$ in descending order, $\mathbf{pk}_{\ell-1}[i]$
- (c) for $0 \leq j \leq \ell - 1$ in ascending order, $\mathbf{pk}_{(\ell+j-1) \bmod \ell}[\ell - j]$.

2. Receive $c_0^b, \dots, c_{\ell-1}^b$ where for $0 \leq x \leq \ell - 1$

$$c_x^b = \langle 1, c_x^b[0], \dots, c_x^b[n-1], \mathbf{s}_x \rangle.$$

Send the following ciphertexts to $\mathcal{A}$ in the order described:

- (a) $\mathbf{s}_{\ell-1}$
 - (b) for $\ell + 1 \leq i \leq n$ in descending order, $h_{\ell-1}[i]$
 - (c) for $0 \leq j \leq \ell - 1$ in ascending order, $c_{(\ell+j-1) \bmod \ell}^b[\ell - j - 1]$.
3. When $\mathcal{A}$ outputs b' , output b' .

When $b = 1$, we observe that

$$\mathbf{s}_{\ell-1} \leftarrow \mathcal{E}(\mathbf{pk}_{\ell-1}[0], \mathbf{sk}_{\ell-1}[n])$$

For $\ell + 1 \leq i \leq n$, (descending)

$$h_{\ell-1}[i] \leftarrow \mathcal{E}(\mathbf{pk}_{\ell-1}[i], \mathbf{sk}_{\ell-1}[i-1])$$

For $0 \leq j \leq \ell - 1$, (ascending)

$$c_{(\ell+j-1) \bmod \ell}^1[\ell - j - 1] \leftarrow \mathcal{E}(\mathbf{pk}_{(\ell+j-1) \bmod \ell}[\ell - j], \mathbf{sk}_{\ell+j \bmod \ell}[\ell - j - 1])$$

which is a $n + 1$ encryption cycle of Π , which means $\mathcal{A}'_\ell$ simulates $\text{CIRC}_{\mathcal{A}, \Pi}(n + 1, \eta)$ when the internal coin of $\text{CIRC}_{\mathcal{A}, \Pi}(n + 1, \eta)$ is 1.

When $b = 0$, we observe that

$$\mathbf{s}_{\ell-1} \leftarrow \mathcal{E}(\mathbf{pk}_{\ell-1}[0], \mathbf{sk}_{\ell-1}[n])$$

For $\ell + 1 \leq i \leq n$, (descending)

$$h_{\ell-1}[i] \leftarrow \mathcal{E}(\mathbf{pk}_{\ell-1}[i], \mathbf{sk}_{\ell-1}[i-1])$$

For $0 \leq j \leq \ell - 1$, (ascending)

$$c_{(\ell+j-1) \bmod \ell}^0[\ell - j - 1] \leftarrow \mathcal{E}(\mathbf{pk}_{(\ell+j-1) \bmod \ell}[\ell - j], 0^{|\mathbf{sk}_0[0]|})$$

and by Lemma 1, this simulates $\text{CIRC}_{\mathcal{A}, \Pi}(n + 1, \eta)$ when the internal coin of $\text{CIRC}_{\mathcal{A}, \Pi}(n + 1, \eta)$ is 0 except with negligible probability.

This implies that if $\epsilon(\eta)$ is not negligible, then Π' is not ℓ -circular secure. Therefore we conclude if Π is $(n + 1)$ -circular insecure and CPA secure, then Π' is $(\leq n)$ -circular insecure. $\square$

4.3 CPA security

Let Π be a CPA secure public key encryption scheme. We show Π' obtained from Π via Construction 1 is a CPA secure public key encryption scheme via a sequence of games [Sho06].

Without loss of generality, we fix an efficient adversary $\mathcal{A}'$ to play in $\text{CPA}_{\mathcal{A}', \Pi'}(\eta)$.

Game 0. This is just the game $\text{CPA}_{\mathcal{A}', \Pi'}(\eta)$ with each step stated explicitly.

1. $(\mathbf{pk}[0], \mathbf{sk}[0]) \leftarrow \mathcal{G}(1^\eta), \dots, (\mathbf{pk}[n], \mathbf{sk}[n]) \leftarrow \mathcal{G}(1^\eta)$.
Compute $\mathbf{s} \leftarrow \mathcal{E}(\mathbf{pk}[0], \mathbf{sk}[n])$, for $3 \leq i \leq n$ compute $h[i] \leftarrow \mathcal{E}(\mathbf{pk}[i], \mathbf{sk}[i-1])$.
Give η and $\mathbf{pk}[0], \dots, \mathbf{pk}[n], \mathbf{s}, h[n], \dots, h[3]$ to $\mathcal{A}'$.
2. $b \leftarrow \$\{0, 1\}$

3. $(m_0, m_1) \leftarrow \mathcal{A}'$.
4. If $m_b \neq |\text{sk}[0], \dots, \text{sk}[n-1]|$, then compute $c^b \leftarrow \langle 0, \mathcal{E}(\text{pk}[1], m_b) \rangle$.
Otherwise,
 - (a) Parse m_b into $m_b[0], \dots, m_b[n-1]$.
 - (b) For $0 \leq i \leq n-1$, compute $c^b[i] \leftarrow \mathcal{E}(\text{pk}[i+1], m_b[i])$.
 Return $c_b = \langle 1, c^b[0], \dots, c^b[n-1], s \rangle$ to $\mathcal{A}'$.
5. $\mathcal{A}'$ outputs b' . The experiment outputs 1 if $b = b'$, 0 otherwise.

Let S_0 denote the event that Game 0. results in 1.

Game 1. Here, we modify how the public key is generated in step 1 of Game 0. Specically, we change how s and $h[n], \dots, h[3]$ are computed in Game 1. Instead of encrypting $\text{sk}[n], \dots, \text{sk}[2]$, they will encrypt strings zeros.

1. $(\text{pk}[0], \text{sk}[0]) \leftarrow \mathcal{G}(1^\eta), \dots, (\text{pk}[n], \text{sk}[n]) \leftarrow \mathcal{G}(1^\eta)$.
Compute $s \leftarrow \mathcal{E}(\text{pk}[0], 0^{|\text{sk}[n]|})$, for $3 \leq i \leq n$ compute $h[i] \leftarrow \mathcal{E}(\text{pk}[i], 0^{|\text{sk}[i-1]|})$.
Give η and $\text{pk}[0], \dots, \text{pk}[n], s, h[n], \dots, h[3]$ to $\mathcal{A}'$.

Let S_1 denote the event that the result of Game 1. is 1.

Claim. For some negligible function negl ,

$$|\Pr[S_0] - \Pr[S_1]| \leq \text{negl}(\eta).$$

Proof. We show this by applying Lemma 1. We define an adversary $\mathcal{A}$ that plays $\text{ZERO}_{\mathcal{A}, \Pi}(n+1, 2, \eta)$ as follows.

$\mathcal{A}$:

1. Receive $\eta, \text{pk}_0, \dots, \text{pk}_n$ and $c_0^b, \dots, c_n^b$.
2. Send the following to $\mathcal{A}'$ in the order described. $\mathcal{A}$ then sends the following in the order described to $\mathcal{A}'$:
 - (a) For $1 \leq i \leq n+1$ in descending order, $\text{pk}_{i+1 \bmod (n+1)}$
 - (b) For $1 \leq i \leq n-1$ in ascending order, c_i^b .

What is being send above, depending on coin b , creates the public key, “ $\text{pk}[0], \dots, \text{pk}[n], s, h[n], \dots, h[3]$ ”, of either Game 0. or of Game 1.

3. Flip a coin d .
4. When $\mathcal{A}'$ queries two equal length challenge messages, m_0, m_1 , if $|m_d| \neq n|\text{sk}_0|$, compute $c^d \leftarrow \mathcal{E}(\text{pk}_1, m_d)$ and send $\langle 0, c^d \rangle$ to $\mathcal{A}'$.
Otherwise parse m_d to $m_d[0], \dots, m_d[n-1]$, and for $0 \leq i \leq n-1$ in ascending order compute

$$c^d[i] \leftarrow \mathcal{E}(\text{pk}_{n+1-i}, m_d[i]).$$

Send $\langle 1, c^d[i], \dots, c^d[n-1], c_1^b \rangle$ to $\mathcal{A}'$.

5. When $\mathcal{A}'$ outputs d' , $\mathcal{A}$ outputs 1 if $d = d'$, output 0 otherwise.

Since each pk_i is generated by $\mathcal{G}$ of Π . We observe that in step 2 of the experiment, $\mathcal{A}'$ receives the following from $\mathcal{A}$:

- If $b = 1$:

$$\text{pk}_1, \text{pk}_0, \text{pk}_n, \dots, \text{pk}_3, \text{pk}_2, \mathcal{E}(\text{pk}_1, \text{sk}_2), \mathcal{E}(\text{pk}_2, \text{sk}_3), \dots, \mathcal{E}(\text{pk}_{n-1}, \text{sk}_n).$$

Which is identical to what is sent to $\mathcal{A}'$ in step 1 of Game 0.

- If $b = 0$:

$$\text{pk}_1, \text{pk}_0, \text{pk}_n, \dots, \text{pk}_3, \text{pk}_2, \mathcal{E}(\text{pk}_1, 0^{|\text{sk}|}), \mathcal{E}(\text{pk}_2, 0^{|\text{sk}|}), \dots, \mathcal{E}(\text{pk}_{n-1}, 0^{|\text{sk}|}).$$

Which is identical to what is sent to $\mathcal{A}'$ in step 1 of Game 1.

Therefore $\mathcal{A}$ simulates Game 0. when $b = 1$ and $\mathcal{A}$ simulates Game 1. when $b = 0$. This means when $b = 1$ $\mathcal{A}$ wins if $\mathcal{A}'$ loses, and when $b = 0$ $\mathcal{A}$ wins if $\mathcal{A}'$ wins. These are the events $\neg S_0$ and S_1 respectively. Therefore, we have the following.

$$\begin{aligned} \Pr[\text{ZERO}_{\mathcal{A}, \Pi}(n, 2, \eta) = 1] &= \frac{1}{2} \Pr[\neg S_0] + \frac{1}{2} \Pr[S_1] \\ &= \frac{1}{2} (1 + \Pr[S_1] - \Pr[S_0]) \\ &= \frac{1}{2} + \frac{1}{2} (\Pr[S_1] - \Pr[S_0]). \end{aligned}$$

Therefore if $|\Pr[S_0] - \Pr[S_1]|$ is not negligible, then by Lemma 1, Π is not CPA secure. $\square$

Now, we define a series of games. The exact number of games depends on the cycle length n in the construction. This n does not depend on the security parameter η . We index a game in this series as Game 2. j for $j \geq 0$.

Game 2. j We let Game 2.0 be Game 1. In Game 2. j for $j > 0$, we modify how the query is answered in step 4 of Game 2.($j - 1$).

4. If $m_b \neq |\text{sk}[0], \dots, \text{sk}[n-1]|$, then compute $c^b \leftarrow \langle 0, \mathcal{E}(\text{pk}[1], m_b) \rangle$. Otherwise,
 - (a) Parse m_b into $m_b[0], \dots, m_b[n-1]$.
 - (b) For $0 \leq i \leq n-j-1$, compute $c^b[i] \leftarrow \mathcal{E}(\text{pk}[i+1], m_b[i])$.
 - (c) For $n-j \leq i \leq n-1$, compute $c^b[i] \leftarrow \mathcal{E}(\text{pk}[i+1], 0^{|\text{sk}[0]|})$.
 Return $c_b = \langle 1, c^b[0], \dots, c^b[n-1] \rangle$ to $\mathcal{A}'$.

For $0 \leq j$, we let $S_{2.j}$ denote the event that Game 2. j . results in 1.

Claim. For some negligible function negl ,

$$|\Pr[S_{2.j}] - \Pr[S_{2.(j+1)}]| \leq \text{negl}(\eta).$$

Proof. We defined an adversary $\mathcal{A}$ to play $\text{CPA}_{\mathcal{A}, \Pi}(\eta)$ as follows.

$\mathcal{A}$:

1. Receive η and pk . Label this pk as pk_{n-j} .
2. For $i \in \{0, \dots, n\} \setminus \{n-j\}$, compute $\text{pk}_i, \text{sk}_i \leftarrow \mathcal{G}(1^\eta)$. Compute $\text{s} \leftarrow \mathcal{E}(\text{pk}_0, 0^{|\text{sk}_0|})$. For $3 \leq i \leq n$, compute $h[i] \leftarrow \mathcal{E}(\text{pk}_i, 0^{|\text{sk}_0|})$. Send

$$\text{pk}_0, \dots, \text{pk}_n, \text{s}, h[n], \dots, h[3]$$
 to $\mathcal{A}'$.
3. Flip a coin d .
4. When $\mathcal{A}$ receive two equal length challenge messages m_0, m_1 from $\mathcal{A}'$, if $|m_d| \neq n \cdot |\text{sk}_0|$, compute $c^d \leftarrow \mathcal{E}(\text{pk}[1], c^d)$ and send $\langle 0, c^d \rangle$ to $\mathcal{A}'$. Else create and query challenge messages $m'_0 = m_d[n - (j+1)]$ and $m'_1 = 0^{|m_d[0]|}$ to receive $c^b[n - (j+1)] \leftarrow \mathcal{E}(\text{pk}_{n-j}, m'_b)$. For $i \in \{0, n-1\} \setminus \{n - (j+1)\}$ compute

$$c^d[i] = \begin{cases} \mathcal{E}(\text{pk}[i+1], m[i]) & \text{if } i \leq n - (j+1) \\ \mathcal{E}(\text{pk}[i+1], 0^{|m[i]|}) & \text{otherwise.} \end{cases}$$

Send $\langle 1, c^d[0], c^d[1], \dots, c^d[n-1], \text{s} \rangle$ to $\mathcal{A}'$.

5. When $\mathcal{A}'$ outputs d' , $\mathcal{A}$ outputs 1 if $d = d'$, output 0 otherwise.

When $b = 0$, $\mathcal{A}$ simulates Game 2. j , and when $b = 1$, $\mathcal{A}$ simulates Game 2. $(j + 1)$ when the internal coin is d . Therefore when $b = 0$, $\mathcal{A}$ wins if $d \neq d'$, which is the probability that $\mathcal{A}'$ fails in Game 2. j . When $b = 1$, $\mathcal{A}$ wins if $\mathcal{A}'$ wins in Game 2. $(j + 1)$. This gives us the following:

$$\begin{aligned} \text{CPA}_{\mathcal{A}, \Pi}(\eta) &= \frac{1}{2} \Pr[\neg S_{2,j}] + \frac{1}{2} \Pr[S_{2,(j+1)}] \\ &= \frac{1}{2} (1 + \Pr[S_{2,(j+1)}] - \Pr[S_{2,j}]). \end{aligned}$$

Therefore, if $|\Pr[S_{2,j}] - \Pr[S_{2,(j+1)}]|$ is not negligible, then Π is not CPA secure. $\square$

Now we show that $\mathcal{A}'$ can only win Game 2. n with negligible probability. This is because the game 2. n is essentially just a CPA game with Π .

Claim. *For some negligible function negl ,*

$$\Pr[S_n] \leq \frac{1}{2} + \text{negl}(\eta).$$

Proof. Notice that in Game 2. n , when the lengths of the challenge messages are not equal to $n|\text{sk}_0|$, then encryptions of zero are returned regardless of the coin of the game. When $\mathcal{A}$ queries challenge messages on other lengths, it receives a ciphertext encrypted by public key pk_1 . Further, in Game 2. n , the key pk_1 is never used as the message of any ciphertext, which means Game 2. n can be simulated by any adversary with a CPA oracle in a CPA game against Π . Therefore, if Π is CPA secure, then

$$\Pr[S_n] \leq \frac{1}{2} + \text{negl}(\eta).$$

$\square$

Lemma 3. *If Π is CPA secure, then Π' is CPA secure.*

Proof. This is shown via the sequence of games above. Assume Π is CPA secure. Let $\text{negl}_0(\eta)$ denote the difference between $\Pr[S_0]$ and $\Pr[S_1]$. Let $\text{negl}_{i+1}(\eta)$ denote the difference between $\Pr[S_{2,i}]$ and $\Pr[S_{2,(i+1)}]$. Let $\text{negl}_{n+1}(\eta)$ denote the advantage of $\mathcal{A}'$ in the CPA game against Π . Then we have

$$\begin{aligned} \text{CPA}_{\mathcal{A}', \Pi'}(\eta) &= \Pr[S_0] \leq \Pr[S_{2,0}] + \text{negl}_0(\eta) \\ &\leq \Pr[S_{2,1}] + \sum_{i \in \{0,1\}} \text{negl}_i(\eta) \\ &\vdots \\ &\leq \Pr[S_{2,n}] + \sum_{i \in \{0,\dots,n\}} \text{negl}_i(\eta) \\ &\leq \frac{1}{2} + \sum_{i \in \{0,\dots,n+1\}} \text{negl}_i(\eta). \end{aligned}$$

This concludes that Π' is CPA secure. $\square$

5 Bit Encryption Construction

In this section, we present a construction that preserves the message space of a base scheme. Construction 2 differs from Construction 1 in the encryption and decryption functions. In Construction 2, we are always expecting the message to have length 1, and we will create n ciphertexts using n of the $n + 1$ public keys.

Construction 2. *Given a CPA secure $(n + 1)$ -circular insecure public key bit encryption scheme $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$. We obtain $\Pi^{\text{bit}} = (\mathcal{G}^{\text{bit}}, \mathcal{E}^{\text{bit}}, \mathcal{D}^{\text{bit}})$ as follows.*

Π^{bit} :

- $\mathcal{G}^{\text{bit}}$: Compute

$$(\text{pk}[0], \text{sk}[0]) \leftarrow \mathcal{G}(1^n), \dots, (\text{pk}[n], \text{sk}[n]) \leftarrow \mathcal{G}(1^n).$$

Compute $s \leftarrow \mathcal{E}(\text{pk}[0], \text{sk}[n])$ (one bit at a time). For $3 \leq i \leq n$ compute $h[i] \leftarrow \mathcal{E}(\text{pk}[i], \text{sk}[i - 1])$ (one bit at a time). Return public key

$$\text{pk} = \text{pk}[0], \dots, \text{pk}[n], s, h[n], h[n - 1], \dots, h[3].$$

and secret key

$$\text{sk} = \text{sk}[0], \dots, \text{sk}[n - 1].$$

- $\mathcal{E}^{\text{bit}}(\text{pk}, m)$: Note that $m \in \{0, 1\}$. Parse pk to

$$\text{pk}[0], \dots, \text{pk}[n], s, h[n], \dots, h[3].$$

For $0 \leq i \leq n - 1$ compute $c[i] \leftarrow \mathcal{E}(\text{pk}[i + 1], m)$, return

$$c[0], \dots, c[n - 1]$$

- $\mathcal{D}^{\text{bit}}(\text{sk}, c)$: Parse c into $c[0], \dots, c[n - 1]$, return

$$\mathcal{D}(\text{sk}[1], c[0]).$$

Construction 2's proof of CPA security is similar to that of Construction 1. We can repeat the sequence of games from Section 4.3 with some modifications. In each encryption, instead of parsing m to n strings each of length $|\text{sk}[0]|$, we will encrypt the message bit m with n 'sub'-public keys. In modified Game 2. n , any key used to encrypt a message does not appear as a message anywhere, and this game is similar to a multiple-key CPA game. A multiple-key CPA game, as its name suggests, is a game played with multiple keys, each encrypting the same challenge message.³ It can be easily observed that an encryption scheme is CPA secure if and only if such a scheme is multiple-key CPA secure.

Before we proceed to show that the encryption scheme is $(\leq n)$ -circular insecure, we provide an example.

Example 1. In this simplified example, we assume $n = 2$, and the secret key of the base scheme Π has length 2. A length 1 encryption cycle of Π^{bit} with $(\text{pk}_0, \text{sk}_0) \leftarrow \mathcal{G}^{\text{bit}}(\eta)$ is the following

$$\begin{aligned} & \mathcal{E}^{\text{bit}}(\text{pk}_0, \text{sk}_0) \\ &= \mathcal{E}^{\text{bit}}(\text{pk}_0, \text{sk}_0[0][0]), \mathcal{E}^{\text{bit}}(\text{pk}_0, \text{sk}_0[0][1]), \mathcal{E}^{\text{bit}}(\text{pk}_0, \text{sk}_0[1][0]), \mathcal{E}^{\text{bit}}(\text{pk}_0, \text{sk}_0[1][1]) \\ &= \mathcal{E}(\text{pk}_0[1], \text{sk}_0[0][0]), \mathcal{E}(\text{pk}_0[2], \text{sk}_0[0][0]), \mathcal{E}(\text{pk}_0[1], \text{sk}_0[0][1]), \mathcal{E}(\text{pk}_0[2], \text{sk}_0[0][1]), \\ & \quad \mathcal{E}(\text{pk}_0[1], \text{sk}_0[1][0]), \mathcal{E}(\text{pk}_0[2], \text{sk}_0[1][0]), \mathcal{E}(\text{pk}_0[1], \text{sk}_0[1][1]), \mathcal{E}(\text{pk}_0[2], \text{sk}_0[1][1]), \end{aligned}$$

³We provide a definition of multiple-key CPA in Appendix B.

where $\mathbf{pk}_0 = \mathbf{pk}_0[0], \mathbf{pk}_0[1], \mathbf{pk}_0[2], \mathbf{s}_0$ and $\mathbf{sk}_0 = \mathbf{sk}_0[0], \mathbf{sk}_0[1]$. For $i, j \in \{0, 1\}$, $\mathbf{sk}_0[i][j]$ denotes the j th bit of $\mathbf{sk}_0[i]$. Note that in this example, we assume $|\mathbf{sk}_0[i]| = 2$. The reason for the expansions above is because Π^{bit} and Π are both bit encryptions, which encrypt messages with lengths longer than 1 bit by bit. If we extract just the underlined terms, we have

$$\mathcal{E}(\mathbf{pk}_0[2], \mathbf{sk}_0[0][0]), \mathcal{E}(\mathbf{pk}_0[2], \mathbf{sk}_0[0][1]) = \mathcal{E}(\mathbf{pk}_0[2])(\mathbf{sk}_0[0]).$$

It can be observed that $\mathbf{s}_0 \leftarrow \mathcal{E}(\mathbf{pk}[0], \mathbf{sk}[2])$. Combining the underlined terms with $\mathbf{s}$ we have $\mathcal{E}(\mathbf{pk}[2], \mathbf{sk}[0]), \mathcal{E}(\mathbf{pk}[0], \mathbf{sk}[2])$ which is a length 2 circular expression of Π .

Circular Insecurity. Now, to show this scheme is $(\leq n)$ -circular insecure. We remind the readers that for bit encryption, when we say $\mathcal{E}(k, m)$ for $|m| > 1$, we mean encrypting each bit of m in the order they appear in m , each time with fresh randomness. Consider an arbitrary $\ell \in \{1, \dots, n\}$. Given $(\mathbf{pk}_0, \mathbf{sk}_0) \leftarrow \mathcal{G}^{\text{bit}}(1^\eta), \dots, (\mathbf{pk}_\ell, \mathbf{sk}_\ell) \leftarrow \mathcal{G}^{\text{bit}}(1^\eta)$, for $0 \leq i < \ell$ let

$$c_i^b \leftarrow \begin{cases} \mathcal{E}^{\text{bit}}(\mathbf{pk}_i, \mathbf{sk}_{i+1 \bmod \ell}) & \text{if } b = 1 \\ \mathcal{E}^{\text{bit}}(\mathbf{pk}_i, 0^{|\mathbf{sk}_0|}) & \text{otherwise.} \end{cases}$$

To keep the notation tidy, let $p = |\mathbf{sk}_0[0]|$.

An adversary $\mathcal{A}_\ell^{\text{bit}}$ in the game $\text{CIRC}_{\mathcal{A}^{\text{bit}}, \Pi^{\text{bit}}}(\ell, \eta)$ after receiving $c_0^b, \dots, c_{\ell-1}^b$ can do the following. First for $0 \leq i < \ell$, parse c_i^b into

$$c_i^b[0], \dots, c_i^b[n-1],$$

where $c_i^b[j]$ for $0 \leq j \leq n-1$ is sampled from $\mathcal{E}^{\text{bit}}(\mathbf{pk}_i, m_b)$ where $m_1 = \mathbf{sk}_{i+1 \bmod \ell}[j]$ and $m_0 = 0^p$.

For $0 \leq j \leq n-1$, $c_i^b[j]$ can be parsed into

$$c_i^b[j][0], c_i^b[j][1], \dots, c_i^b[j][p-1]$$

where for $0 \leq x \leq p$, $c_i^b[j][x]$ is samples from $\mathcal{E}^{\text{bit}}(\mathbf{pk}_i, m'_b)$ where $m'_1 = \mathbf{sk}_{i+1 \bmod \ell}[j][x]$ (the x th bit of $\mathbf{sk}_{i+1 \bmod \ell}[j]$) and $m'_0 = 0$.

For $0 \leq x \leq p-1$, $c_i^b[j][x]$ can be parsed to

$$c_i^b[j][x][0], \dots, c_i^b[j][x][n-1]$$

where for $0 \leq y \leq n-1$, $c_i^b[j][x][y]$ is sampled from $\mathcal{E}(\mathbf{pk}_i[y+1], m''_b)$ where $m''_1 = \mathbf{sk}_{i+1 \bmod \ell}[j][x]$ and $m''_0 = 0$.

Remark 2. Putting what we have just described into Example 1, we have

$$\begin{aligned} c_0^1 &\leftarrow \mathcal{E}^{\text{bit}}(\mathbf{pk}_0, \mathbf{sk}_0) \\ c_0^1[0] &\leftarrow \mathcal{E}^{\text{bit}}(\mathbf{pk}_0, \mathbf{sk}_0[0]) \\ c_0^1[1] &\leftarrow \mathcal{E}^{\text{bit}}(\mathbf{pk}_0, \mathbf{sk}_0[1]) \\ c_0^1[0][0] &\leftarrow \mathcal{E}^{\text{bit}}(\mathbf{pk}_0, \mathbf{sk}_0[0][0]) = \mathcal{E}(\mathbf{pk}_0[1], \mathbf{sk}_0[0][0]), \mathcal{E}(\mathbf{pk}_0[2], \mathbf{sk}_0[0][0]) \\ c_0^1[0][1] &\leftarrow \mathcal{E}^{\text{bit}}(\mathbf{pk}_0, \mathbf{sk}_0[0][1]) = \mathcal{E}(\mathbf{pk}_0[1], \mathbf{sk}_0[0][1]), \mathcal{E}(\mathbf{pk}_0[2], \mathbf{sk}_0[0][1]) \\ \underline{c_0^1[0][0][1]} &\leftarrow \mathcal{E}(\mathbf{pk}_0[2], \mathbf{sk}_0[0][0]) \\ \underline{c_0^1[0][1][1]} &\leftarrow \mathcal{E}(\mathbf{pk}_0[2], \mathbf{sk}_0[0][1]) \end{aligned}$$

The underlined terms correspond to the underlined terms in Example 1.

Now that the ciphertext has been parsed, the adversary can extract the following in the order described:

1. $s_{\ell-1}$
2. for $\ell+1 \leq i \leq n$ in descending order $h_{\ell-1}[i]$
3. for $0 \leq j \leq \ell-1$ in ascending order:
 - (a) for $0 \leq x \leq p-1$ in ascending order: $c_{(\ell+j-1) \bmod \ell}^b[\ell-j-1][x][\ell-j-1]$.

Since

$$\begin{aligned}
 & c_{(\ell+j-1) \bmod \ell}^b[\ell-j-1][0][\ell-j-1], \dots, c_{(\ell+j-1) \bmod \ell}^b[\ell-j-1][p-1][\ell-j-1] \\
 & \leftarrow \mathcal{E}(\mathbf{pk}_{(\ell+j-1) \bmod i}[\ell-j], \mathbf{sk}_{(\ell+j) \bmod \ell}[\ell-j-1][0]), \\
 & \quad \vdots \\
 & \mathcal{E}(\mathbf{pk}_{(\ell+j-1) \bmod i}[\ell-j], \mathbf{sk}_{(\ell+j) \bmod \ell}[\ell-j-1][p-1]) \\
 & = \mathcal{E}(\mathbf{pk}_{(\ell+j-1) \bmod i}[\ell-j], \mathbf{sk}_{(\ell+j) \bmod \ell}[\ell-j-1]).
 \end{aligned}$$

The ciphertexts that the adversary has extracted when $b = 1$ are sampled from:

1. $s_{\ell_1} \leftarrow \mathcal{E}(\mathbf{pk}_{\ell}[0], \mathbf{sk}_{\ell}[n])$
2. for $\ell+1 \leq i < n$ in descending order, $h_{\ell-1}[i] \leftarrow \mathcal{E}(\mathbf{pk}_{\ell-1}[i], \mathbf{sk}_{\ell-1}[i-1])$
3. for $0 \leq j \leq \ell-1$ in ascending order,

$$\begin{aligned}
 & c_{(\ell+j-1) \bmod \ell}^1[\ell-1-j][0, \dots, p-1][\ell-j-1] \\
 & \leftarrow \mathcal{E}(\mathbf{pk}_{(\ell+j-1) \bmod i}[\ell-j], \mathbf{sk}_{(\ell+1) \bmod \ell}[\ell-j-1]).
 \end{aligned}$$

which is a $n+1$ -cycle with the following keys in Π ,

1. $\mathbf{pk}_{\ell-1}[0], \mathbf{sk}_{\ell-1}[0]$
2. for $\ell+1 \leq i < n$, $\mathbf{pk}_{\ell-1}[i], \mathbf{sk}_{\ell-1}[i]$
3. for $0 \leq j \leq \ell-1$, $\mathbf{pk}_{(\ell+j-1) \bmod \ell}[\ell-j], \mathbf{sk}_{(\ell+j-1) \bmod \ell}[\ell-j]$.

And in the event where $b = 0$, the ciphertexts extracted by $\mathcal{A}^{\text{bit}}$ are indistinguishable from encryptions of zeros since it is only “almost a cycle” as in Lemma 1.

5.1 Arbitrary Message Space

To modify an $(n+1)$ -circular insecure encryption scheme Π with message length β using Construction 2 and preserve the same message space, we first define a scheme $\bar{\Pi}$ where each key is padded with extra bits at the end so that the length of a key in $\bar{\Pi}$ is divisible by β . This step depends on how the encryption function of the base scheme pads the plaintext when it is shorter than its message space. Now, with the appropriate changes for calls to the encryption function of the base scheme, by encrypting β bits at a time using the same n keys as the bit version, Construction 2 is ready to be used.

6 Symmetric Encryption Scheme

This section discusses how to modify the constructions to work in the symmetric key setting. There are three changes to be made. The first change is to handle the fact that the public key is the secret key. The key generation function outputs all $n+1$ base secret keys. The terms $\mathbf{s}$ and $h[\dots]$ are no longer computed by the key generation function;

now they will be computed by the encryption function. The second change requires the encryption function to perform the computations of s and $h[n], \dots, h[3]$ and append the results as part of the ciphertext. i.e., each time the encryption function computes $s \leftarrow \mathcal{E}(\text{sk}[0], \text{sk}[n]), h[n] \leftarrow \mathcal{E}(\text{sk}[n], \text{sk}[n-1]), \dots, h[3] \leftarrow \mathcal{E}(\text{sk}[3], \text{sk}[2])$ and appends these ciphertexts at the end of the ciphertext of the message. The third is, if using Construction 1, during encryption, when expecting a message of the same length as the key, the function needs to perform an extra encryption to accommodate the new n -th key (zero-indexing). This encryption can be performed using sub-key $\text{sk}[1]$.

We show what Construction 1 would look like in the symmetric key setting. Let $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$ be a CPA secure and $(n+1)$ -circular insecure. We can define $\Pi^{\text{sym}} = (\mathcal{G}^{\text{sym}}, \mathcal{E}^{\text{sym}}, \mathcal{D}^{\text{sym}})$.

- $\mathcal{G}^{\text{sym}}(1^n)$: $\text{sk}[0] \leftarrow \mathcal{G}(1^n), \dots, \text{sk}[n] \leftarrow \mathcal{G}(1^n)$. Return $\text{sk}^{\text{sym}} = \text{sk}[0], \dots, \text{sk}[n]$.
- $\mathcal{E}^{\text{sym}}(\text{sk}^{\text{sym}}, m)$: if $|m| \neq |\text{sk}^{\text{sym}}|$, then return $\langle 0, \mathcal{E}(\text{sk}[1], m) \rangle$. Else, parse m to $m[0], \dots, m[n]$. Let $s, h[n], \dots, h[3]$ be $\mathcal{E}(\text{sk}[0], \text{sk}[n]), \mathcal{E}(\text{sk}[n], \text{sk}[n-1]), \dots, \mathcal{E}(\text{sk}[3], \text{sk}[2])$. Return $\langle 1, \mathcal{E}(\text{sk}[1], m[0]), \dots, \mathcal{E}(\text{sk}[n], m[n-1]), \mathcal{E}(\text{sk}[1], m[n]), s, h[n], \dots, h[3] \rangle$.
- $\mathcal{D}^{\text{sym}}(\text{sk}^{\text{sym}}, \langle d, c' \rangle)$: if $d = 0$, return $\mathcal{D}(\text{sk}[1], m)$. Else parse c' to $c[0], \dots, c[n]$, $s, h[n], \dots, h[3]$. Return

$$\mathcal{D}(\text{sk}[1], c[0]) \parallel \mathcal{D}(\text{sk}[2], c[1]) \parallel \dots \parallel \mathcal{D}(\text{sk}[n], c[n-1]) \parallel \mathcal{D}(\text{sk}[1], c[n]).$$

The proof of security is not too different from the public key setting. The only difference is in Game 2.1, we are replacing both $c[0]$ and $c[n]$ with encryptions of zeros. To show the probability that Game 2.1 results in 1 is close to the probability that Game 2.0 or Game 2.2 result in 1, we can reduce the difference between the probabilities into the advantage of an adversary winning left-right oracle game ⁴ against the base encryption scheme Π . It is well known that an encryption scheme is LR-oracle secure if and only if it is CPA secure.

Acknowledgement

I thank Bruce Kapron for the helpful comments.

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⁴See LR-oracle experiment in [KL07].

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A Folklore Constructions for Length 1 Cycle

Let $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$ be a CPA-secure encryption scheme. Let f be a one-way function.

We present folklore constructions in the public key setting, which can be modified minimally to work in the symmetric key setting. We assume Π 's message space is equal to its key space. When the message space is larger, we can pad the secret keys with random bits to achieve the same space. Our presentation may differ from the presentations of others. Still, the core idea of the folklore construction is for the encryption function to “check” if the message is the secret key to the public key.

We can build an 1-circular secure encryption scheme $\Pi^{1\text{-sec}} = (\mathcal{G}^{1\text{-sec}}, \mathcal{E}^{1\text{-sec}}, \mathcal{D}^{1\text{-sec}})$ from Π as follows:

- $\mathcal{G}^{1\text{-sec}}(1^\eta)$: $(pk_0, sk_0) \leftarrow \mathcal{G}(1^\eta)$, $(pk_1, sk_1) \leftarrow \mathcal{G}(1^\eta)$.
Let $pk^{1\text{-sec}} = (f(sk_1), pk_0)$ and $sk^{1\text{-sec}} = (sk_0, sk_1)$.
Return $pk^{1\text{-sec}}$ and $sk^{1\text{-sec}}$.
- $\mathcal{E}^{1\text{-sec}}(pk^{1\text{-sec}}, m)$: Parse $pk^{1\text{-sec}}$ to $(f(sk_1), pk_0)$.
Parse m to $(m[0], m[1])$ where $|m[0]| = |sk_0|$ and $|m[1]| = |sk_1|$.
If $f(m[1]) = f(sk_1)$, return $(\mathcal{E}(pk_0, 1), \mathcal{E}(pk, r))$ where $r \leftarrow \$\{0, 1\}^{|m|}$.
Otherwise, return $(\mathcal{E}(pk_0, 0), \mathcal{E}(pk_0, m))$.
- $\mathcal{D}^{1\text{-sec}}(sk^{1\text{-sec}}, c)$: Parse c to (c_0, c_1) and $sk^{1\text{-sec}}$ to (sk_0, sk_1) .
If $\mathcal{D}(sk_0, c_0) = 1$, return (sk_0, sk_1) .
Otherwise, return $\mathcal{D}(sk_0, c_1)$.

This encryption scheme has “overwhelming correctness”.

To create a 1-circular insecure encryption, we can create $\Pi^{1\text{-ins}} = (\mathcal{G}^{1\text{-ins}}, \mathcal{E}^{1\text{-ins}}, \mathcal{D}^{1\text{-ins}})$:

- $\mathcal{G}^{1\text{-ins}}(1^\eta)$: $(pk_0, sk_0) \leftarrow \mathcal{G}(1^\eta)$, $(pk_1, sk_1) \leftarrow \mathcal{G}(1^\eta)$.
Let $pk^{1\text{-ins}} = (f(sk_1), pk_0)$ and $sk^{1\text{-ins}} = (sk_0, sk_1)$.
Return $pk^{1\text{-ins}}$ and $sk^{1\text{-ins}}$.
- $\mathcal{E}^{1\text{-ins}}(pk^{1\text{-ins}}, m)$: Parse $pk^{1\text{-ins}}$ to $(f(sk_1), pk_0)$.
Parse m into $(m[0], m[1])$.
If $f(m[1]) = f(sk_1)$, return $(1, \mathcal{E}(pk_0, m))$.
Otherwise, return $(0, \mathcal{E}(pk_0, m))$.
- $\mathcal{D}^{1\text{-ins}}(sk^{1\text{-ins}}, c)$: Parse c to (b, c') and $sk^{1\text{-ins}}$ to (sk_0, sk_1) .
Return $\mathcal{D}(sk_0, c')$.

B Multiple-Key CPA security

In the following, we assume n is a non-zero polynomial of η .

Definition 4 (Multiple-Key CPA). Let $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$ be a public key encryption scheme. The n -key CPA experiment for Π and adversary $\mathcal{A}$ denoted by $\text{multi-CPA}_{\mathcal{A}, \Pi}(\eta, n)$ is defined as follows:

$\text{multi-CPA}_{\mathcal{A}, \Pi}(\eta, n)$:

1. $(\text{pk}_0, \text{sk}_0) \leftarrow \mathcal{G}(1^\eta), \dots, (\text{pk}_{n-1}, \text{sk}_{n-1}) \leftarrow \mathcal{G}(1^\eta)$. Give η and $\text{pk}_0, \dots, \text{pk}_n$ to $\mathcal{A}$.
2. $b \leftarrow \$\{0, 1\}$.
3. $\mathcal{A}$ outputs two equal length challenge messages m_0, m_1 .
4. Compute $c_i^b \leftarrow \mathcal{E}(\text{pk}_i, m_b)$ for $i \in \{0, \dots, n-1\}$ and return $c_0^b, \dots, c_{n-1}^b$ to $\mathcal{A}$.
5. $\mathcal{A}$ outputs b' and the experiment results in 1 if $b = b'$, 0 otherwise.

We say Π is n -key CPA secure if for every PPT adversary $\mathcal{A}$,

$$\Pr[\text{multi-CPA}_{\mathcal{A}, \Pi}(\eta, n) = 1] \leq \frac{1}{2} + \text{negl}(\eta).$$

C Details Figure 2 and 3

Consider $n = 2$. Let Π be a 3-circular insecure and CPA secure public key encryption scheme, and suppose Π' is constructed from Π via Construction 1. We can show that Π' is both 1-circular insecure and 2-circular insecure.

Fix a security parameter η . Consider some adversary $\mathcal{A}$ such that

$$\Pr[\text{CIRC}_{\mathcal{A}, \Pi}(3, \eta) = 1] = \frac{1}{2} + \epsilon(\eta).$$

To see Π' is 1-circular insecure, let $(\text{pk}, \text{sk}) \leftarrow \mathcal{G}'(1^\eta)$. Define adversary $\mathcal{A}'_1$ as follows:

$\mathcal{A}'_1$:

1. Receive η and $\text{pk} = \text{pk}[0], \text{pk}[1], \text{pk}[2], \text{s}$. Send $\text{pk}[0], \text{pk}[2], \text{pk}[1]$ to $\mathcal{A}$.
2. Receive $c^b = \langle 1, c^b[0], c^b[1] \rangle$. Send $\text{s}, c^b[1], c^b[0]$ to $\mathcal{A}$.
3. When $\mathcal{A}$ outputs b' , output b' .

In the event $b = 1$, $\mathcal{A}'_1$ has sent ciphertexts

$$\text{s} \leftarrow \mathcal{E}(\text{pk}[0], \text{sk}[2]) \quad c^1[1] \leftarrow \mathcal{E}(\text{pk}[2], \text{sk}[1]) \quad c^1[0] \leftarrow \mathcal{E}(\text{pk}[1], \text{sk}[0])$$

to $\mathcal{A}$, which simulates $\text{CIRC}_{\mathcal{A}, \Pi}(3, \eta)$ when its internal coin is 1.

In the event $b = 0$, $\mathcal{A}'_1$ has sent ciphertexts

$$\text{s} \leftarrow \mathcal{E}(\text{pk}[0], \text{sk}[2]) \quad c^0[1] \leftarrow \mathcal{E}(\text{pk}[2], 0^{|\text{sk}[1]|}) \quad c^0[0] \leftarrow \mathcal{E}(\text{pk}[1], 0^{|\text{sk}[0]|})$$

to $\mathcal{A}$, which by Lemma 1, simulates $\text{CIRC}_{\mathcal{A}, \Pi}(3, \eta)$ when its internal coin is 0 except with negligible probability. Therefore we conclude that $\text{CIRC}_{\mathcal{A}'_1, \Pi'}(1, \eta) = \text{CIRC}_{\mathcal{A}, \Pi}(3, \eta)$ except with negligible probability.

To see Π' is 2-circular secure. Consider $(\text{pk}_0, \text{sk}_0) \leftarrow \mathcal{G}'(1^\eta), (\text{pk}_1, \text{sk}_1) \leftarrow \mathcal{G}'(1^\eta)$. Define adversary $\mathcal{A}'_2$ as follows:

$\mathcal{A}'_2$:

1. Receive η and $\mathbf{pk}_0 = \mathbf{pk}_0[0], \mathbf{pk}_0[1], \mathbf{pk}_0[2], s_0$ and $\mathbf{pk}_1 = \mathbf{pk}_1[0], \mathbf{pk}_1[1], \mathbf{pk}_1[2], s_1$. Send $\mathbf{pk}_1[0], \mathbf{pk}_1[2], \mathbf{pk}_0[1]$ to $\mathcal{A}$.
2. Receive $c_0^b = \langle 1, c_0^b[0], c_0^b[1] \rangle$ and $c_1^b = \langle 1, c_1^b[0], c_1^b[1] \rangle$. Send $s_1, c_1^b[1], c_0^b[0]$ to $\mathcal{A}$.
3. When $\mathcal{A}$ outputs b' , output b' .

In the event $b = 1$, $\mathcal{A}'_2$ has sent ciphertexts

$$s_1 \leftarrow \mathcal{E}(\mathbf{pk}_1[0], \mathbf{sk}_1[2]) \quad c_1^1[1] \leftarrow \mathcal{E}(\mathbf{pk}_1[2], \mathbf{sk}_0[1]) \quad c_0^1[0] \leftarrow \mathcal{E}(\mathbf{pk}_0[1], \mathbf{sk}_1[0])$$

to $\mathcal{A}$, which simulates $\text{CIRC}_{\mathcal{A}, \Pi}(3, \eta)$ when its internal coin is 1.

In the event $b = 0$, $\mathcal{A}'_2$ has sent ciphertexts

$$s_1 \leftarrow \mathcal{E}(\mathbf{pk}_1[0], \mathbf{sk}_1[2]) \quad c_1^0[1] \leftarrow \mathcal{E}(\mathbf{pk}_1[2], 0^{|\mathbf{sk}_0[1]|}) \quad c_0^0[0] \leftarrow \mathcal{E}(\mathbf{pk}_0[1], 0^{|\mathbf{sk}_1[0]|})$$

By Lemma 1 we conclude that $\text{CIRC}_{\mathcal{A}'_2, \Pi'}(2, \eta) = \text{CIRC}_{\mathcal{A}, \Pi}(3, \eta)$ except with negligible probability.

D Circular Security

Let $\Pi = (\mathcal{G}, \mathcal{E}, \mathcal{D})$ be a $n + 1$ -circular secure CPA secure PKE.

We define $\Pi' = (\mathcal{G}', \mathcal{E}', \mathcal{D}')$. Π' has message space $n \times \{0, 1\}^{|\mathbf{sk}|}$ where $\mathbf{sk} \leftarrow \mathcal{G}(1^\eta)$.

- $\mathcal{G}'(1^\eta)$:

1. $(\mathbf{pk}[0], \mathbf{sk}[0]), \dots, (\mathbf{pk}[n], \mathbf{sk}[n]) \leftarrow \mathcal{G}(1^\eta)$.
2. $s \leftarrow \mathcal{E}(\mathbf{pk}_{n-1}, \mathbf{sk}_n)$.
3. Return public key $\mathbf{pk}'$:

$$\mathbf{pk}[0], \dots, \mathbf{pk}[n-2], \mathbf{pk}[n], s$$

return secret key $\mathbf{sk}'$:

$$\mathbf{sk}[0], \dots, \mathbf{sk}[n-1]$$

- $\mathcal{E}'(\mathbf{pk}', m)$:

1. Parse m into $m[0], \dots, m[n-1]$.
2. Parse $\mathbf{pk}'$ into $\mathbf{pk}[0], \dots, \mathbf{pk}[n-2], \mathbf{pk}[n]$
3. for $i \in [n+1]$ compute $c[i] \leftarrow \begin{cases} \mathcal{E}(\mathbf{pk}[i], m[i+1 \bmod (n+1)]) & i \neq n-1 \\ s & i = n-1 \end{cases}$
4. Return $c[0], \dots, c[n]$.

- $\mathcal{D}'(\mathbf{sk}', c')$:

1. parse $c' = c'[0], \dots, c'[n]$:
2. parse $\mathbf{sk}' = \mathbf{sk}[0], \dots, \mathbf{sk}[n-1]$.
3. $\mathbf{sk}[n] \leftarrow \mathcal{D}(\mathbf{sk}[n-1], c[n-1])$
4. For $i \in [n]$, compute $m[i] = \mathcal{D}(\mathbf{sk}[i-1 \bmod n+1], c[i-1 \bmod n+1])$.
5. Return $m[0] \parallel \dots \parallel m[n-1]$

n -circular security: A length n encryption cycle of Π' will consist of exactly n length $n+1$ encryption cycles of Π . Each cycle will not contain any keys from other cycles.